

TypeScript -1

Why TypeScript?

- JavaScript is a dynamic programming language with no type system.
- JavaScript provides primitive types like string, number, object, etc., but it doesn't check assigned values.
- JavaScript variables are declared using the var keyword, and it can point to any value.

Why TypeScript?

- JavaScript doesn't support classes and other object-oriented features (ECMA2015 supports it).
- So, without the type system, it is not easy to use JavaScript to build complex applications with large teams working on the same code.

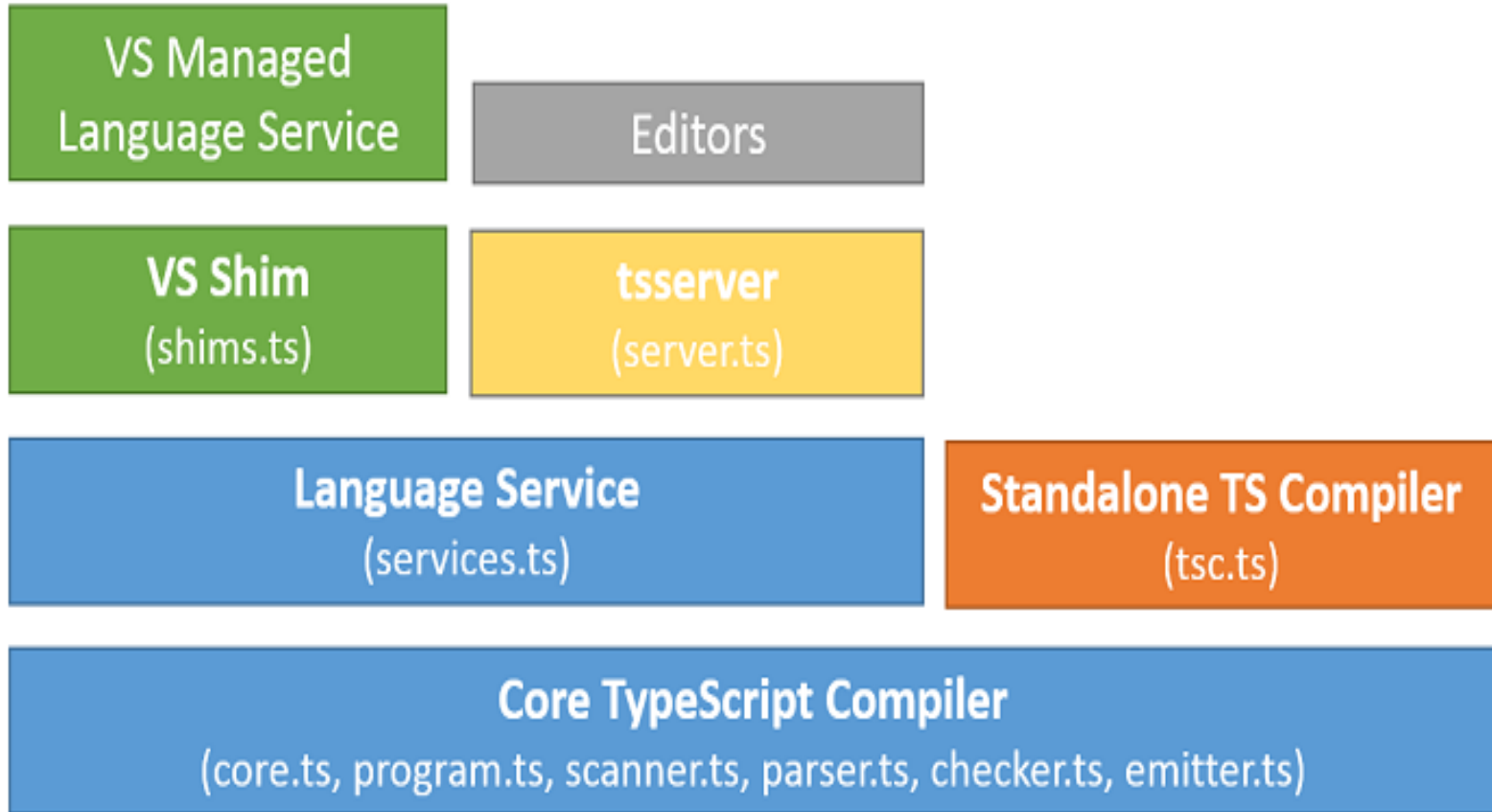
Why TypeScript?

- The type system increases the code quality, readability and makes it easy to maintain. More importantly, errors can be caught at compile time rather than at runtime.
- Hence, the reason to use TypeScript is that it catches errors at compile-time, so that you can fix it before you run code.

Why TypeScript?

- It supports object-oriented programming features like data types, classes, enums, etc., allowing JavaScript to be used at scale.
- TypeScript compiles into simple JavaScript. The TypeScript compiler is also implemented in TypeScript and can be used with any browser or JavaScript engines like [Node.js](#). TypeScript needs an ECMAScript 3 or higher compatible environment to compile.

TypeScript Internal Architecture



Main components of the TypeScript architecture

TypeScript Internal Architecture

Components of the TypeScript architecture

- The core TypeScript API is the foundation of everything, and is composed of elements such as the scanner, parser, binder, type checker, and emitter.
- The language services and the standalone compiler (the tsc command-line tool) sit on top of the core compiler APIs.
- Finally, the Visual Studio shim and the TypeScript standalone server (tsserver) are designed to facilitate the integration of TypeScript with Visual Studio and other source code editors.

TypeScript Environment Setup

Local Environment Setup

- Typescript is an Open Source technology. It can run on any browser, any host, and any OS. You will need the following tools to write and test a Typescript program –

TypeScript Environment Setup

A Text Editor

- The text editor helps you to write your source code. Examples of a few editors include Windows Notepad, Notepad++, Emacs, vim or vi, etc. Editors used may vary with Operating Systems.
- The source files are typically named with the extension **.ts**

TypeScript Environment Setup

The TypeScript Compiler

- The TypeScript compiler is itself a **.ts** file compiled down to JavaScript (.js) file. The TSC (TypeScript Compiler) is a source-to-source compiler (transcompiler / transpiler).



TypeScript Environment Setup

- The TSC generates a JavaScript version of the **.ts** file passed to it. In other words, the TSC produces an equivalent JavaScript source code from the Typescript file given as an input to it. This process is termed as transpilation.
- However, the compiler rejects any raw JavaScript file passed to it. The compiler deals with only **.ts** or **.d.ts** files.

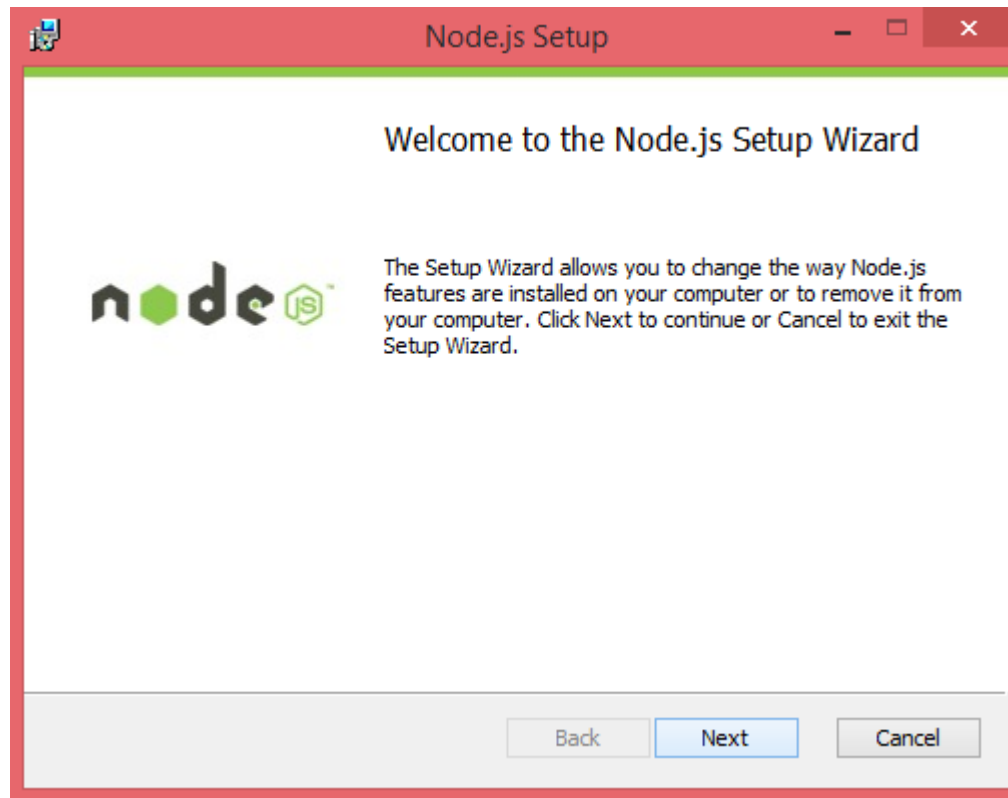
Installing Node.js

- Node.js is an open source, cross-platform runtime environment for server-side JavaScript.
- Node.js is required to run JavaScript without a browser support.
- It uses Google V8 JavaScript engine to execute code.
- You may download Node.js source code or a pre-built installer for your platform. Node is available here – <https://nodejs.org/en/download>

Installation on Windows

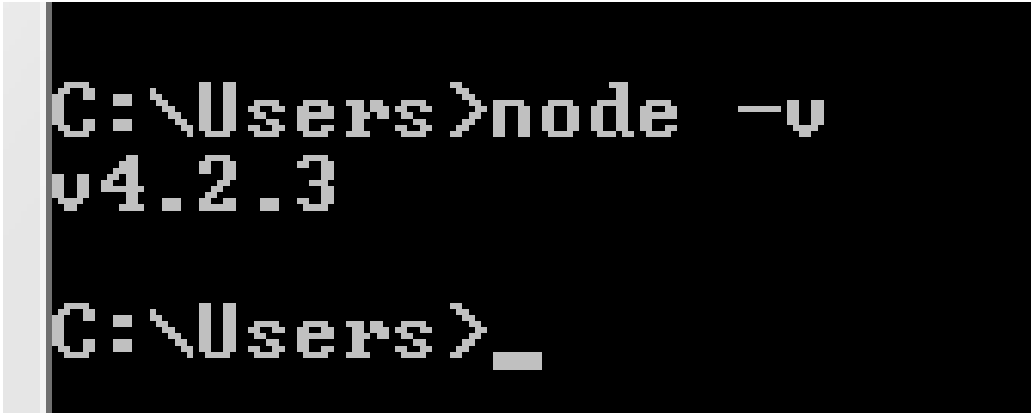
Follow the steps given below to install Node.js in Windows environment.

- **Step 1** – Download and run the .msi installer for Node.



Installation on Windows

- **Step 2** – To verify if the installation was successful, enter the command ***node -v*** in the terminal window.

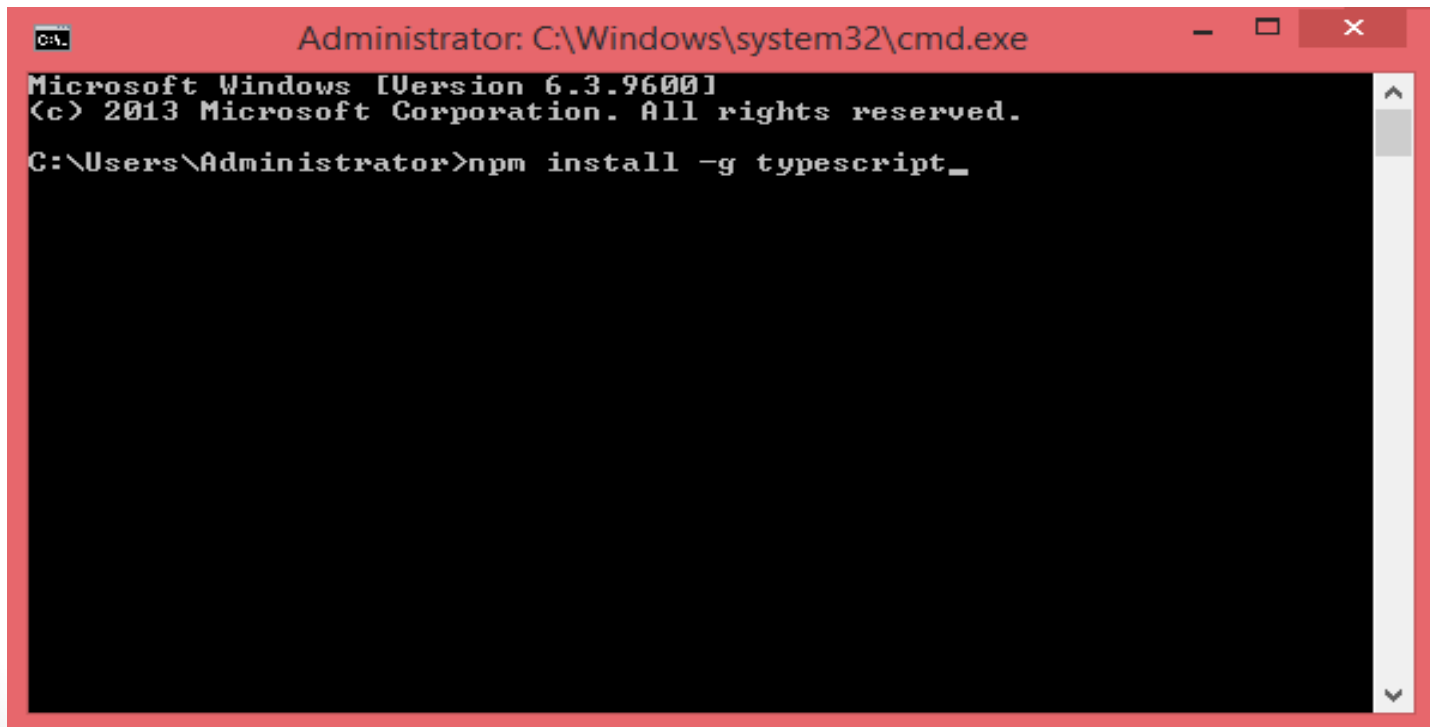


```
C:\Users>node -v
v4.2.3
C:\Users>_
```

Installation on Windows

- **Step 3** – Type the following command in the terminal window to install TypeScript.

`npm install -g typescript`

A screenshot of a Windows command prompt window. The title bar is red and contains the text "Administrator: C:\Windows\system32\cmd.exe". The window has standard Windows window controls (minimize, maximize, close) on the right. The command prompt shows the following text: "Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>npm install -g typescript_". The text is white on a black background. A vertical scrollbar is visible on the right side of the command prompt area.

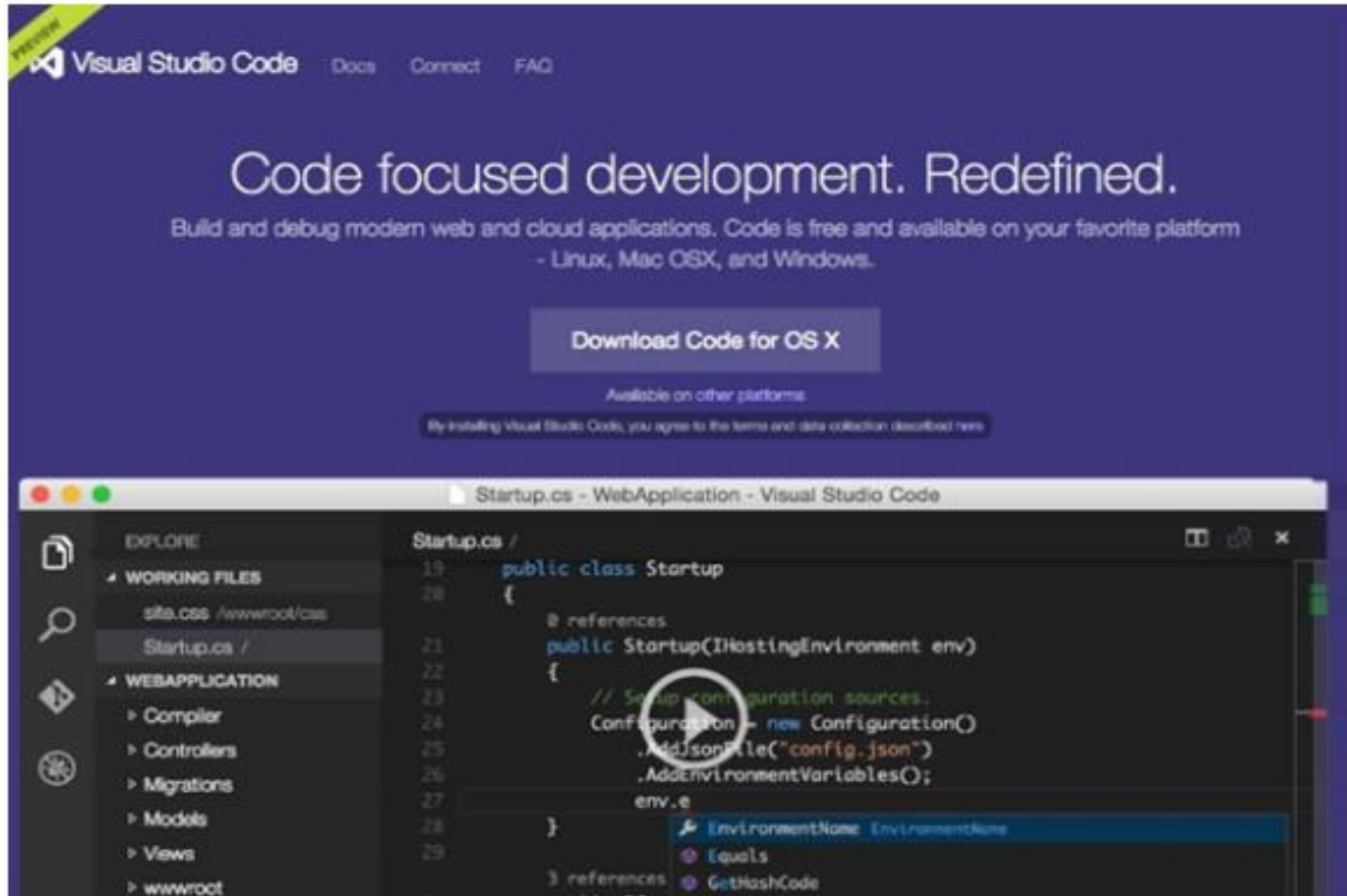
```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.
C:\Users\Administrator>npm install -g typescript_
```

Visual Studio Code

- This is an open source IDE from Visual Studio.
- It is available for Mac OS X, Linux and Windows platforms.
- VScode is available at
 - <https://code.visualstudio.com/>

Installation on Windows

Step 1 – Download Visual Studio Code for Windows.

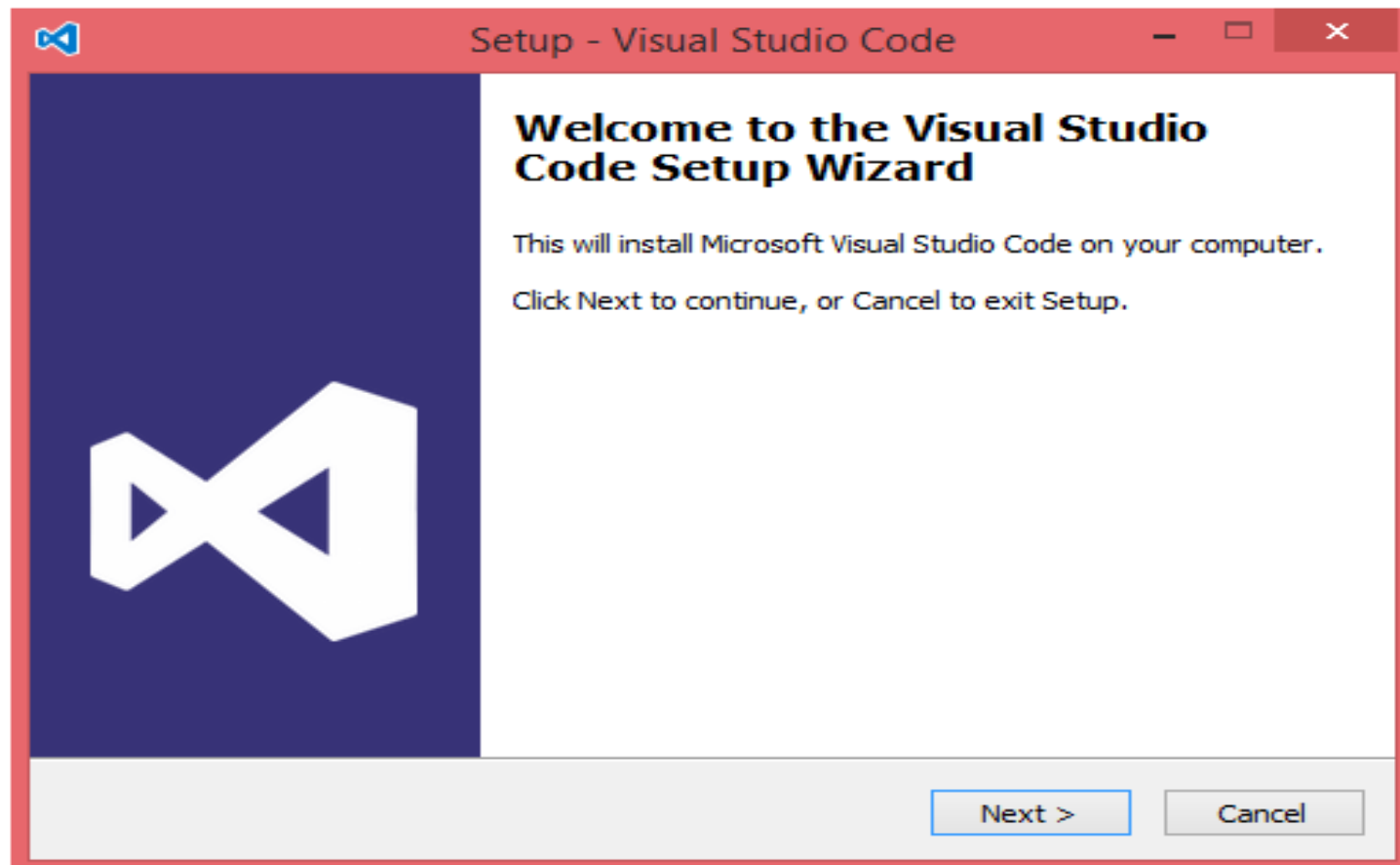


Installation on Windows

Step 2 – Double-click on VSCodeSetup.exe

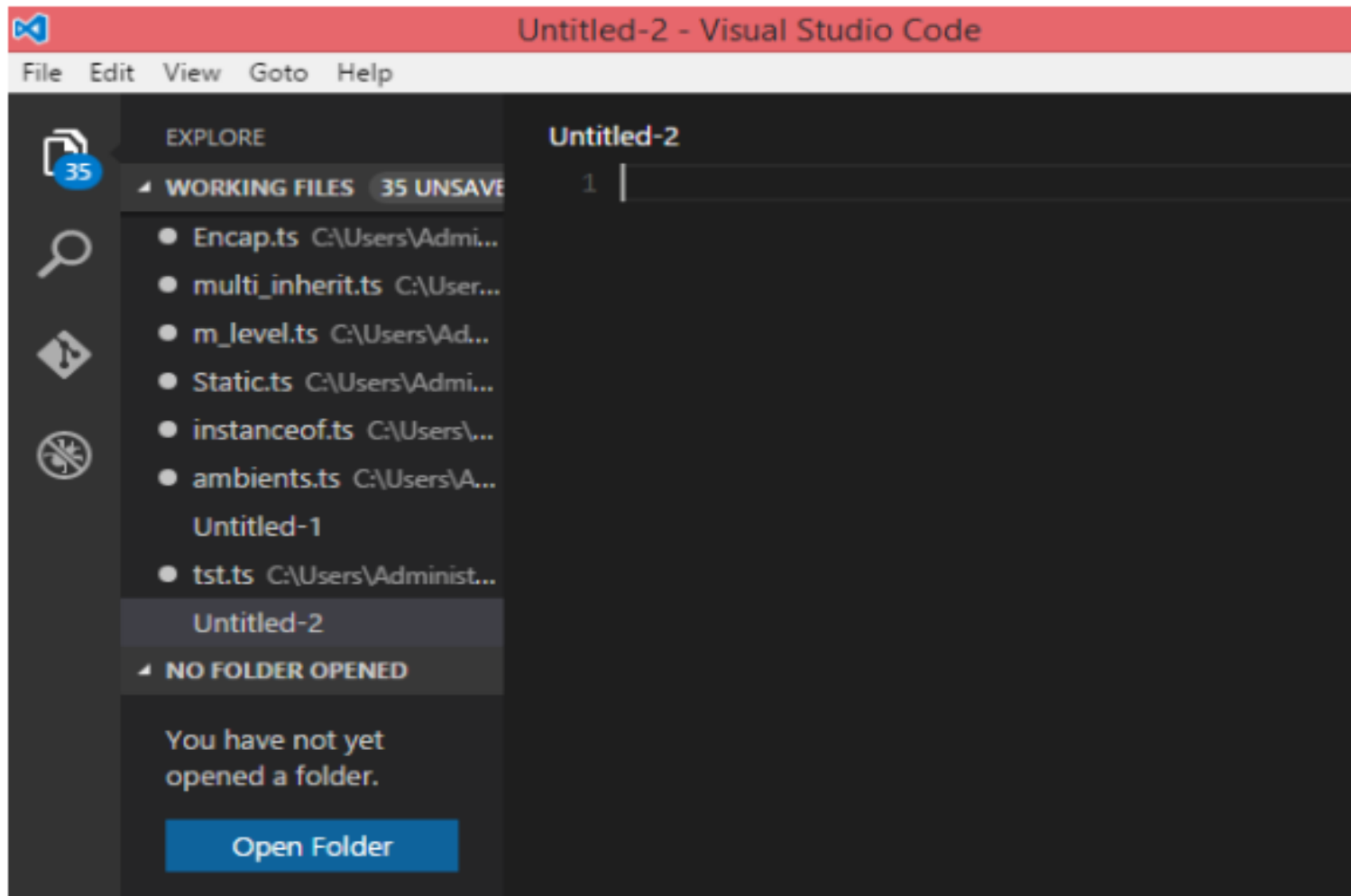


to launch the setup process. This will only take a minute.



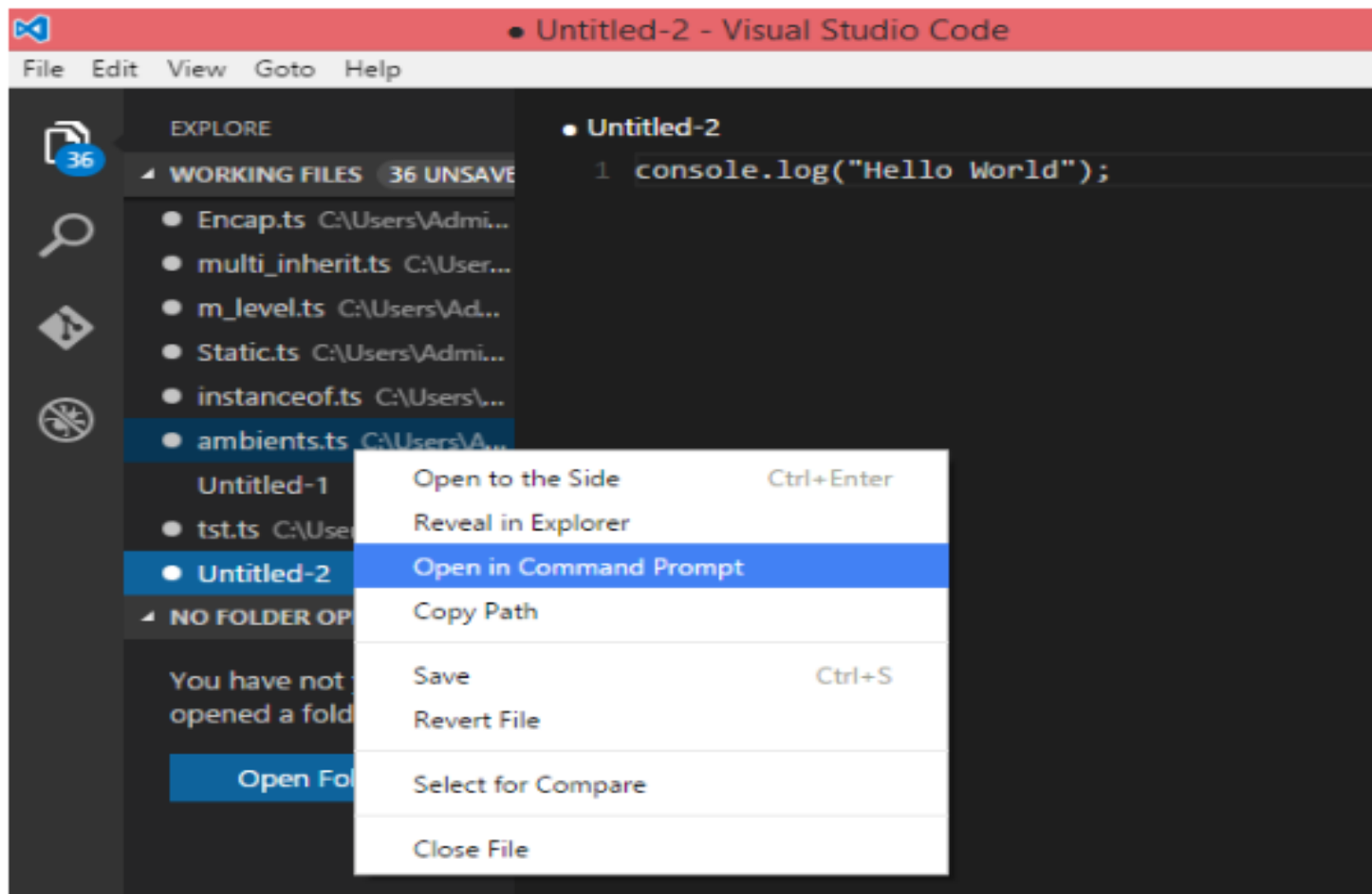
Installation on Windows

Step 3 – A screenshot of the IDE is given below.



Installation on Windows

Step 4 – You may directly traverse to the file's path by right clicking on the file → open in command prompt. Similarly, the Reveal in Explorer option shows the file in the File Explorer.



Thank
you!